Table 2: R^2 values for predicting ROME edit success in Error Injection, subsetted to 10% of the data that has the most concentrated tracing effects in a small number of layers. Even when facts appear to be stored at a small number of layers *and not other layers*, tracing effects are still not predictive of editing performance.

Concentrated Data	R^2 Values		
Method	Layer	Tracing Effect	Both
ROME	0.927	0.02	0.929

pass until a certain set of layers (window size=5), where we zero out representation values for the MLP output representations at the subject token indices within those layers (then continue the forward pass). The localization effect is computed as the proportion of the original predicted probability that is deleted via the zeroing operation (ranging from no effect as 0% to 100% of probability deleted as 100%). These new results are shown in Figs. 21 for rewrite score and 22 for overall score, using ROME on GPT-J with CounterFact data. We obtain the same conclusions as our analysis with causal tracing: localization via representation zeroing is not predictive of edit success. Specifically, we see correlations between edit success and localization effect to be near zero across layers (using either rewrite score or overall score for edit success).

Highly concentrated tracing effects. Since Causal Tracing analysis suggests that information accrues gradually across layers (see Fig. 10), it seems possible that information is simply so diffusely spread across model layers that no matter what layer you edit, you will be editing a layer where a fact is at least stored in part. Based on this observation, we want to test whether tracing effects correlate better with edit success specifically when tracing effects are concentrated in a small number of layers. This condition represents that a fact appears to be stored in a small number of layers *and not elsewhere*. We hope that by editing in that range of layers, we can more easily manipulate that fact. To identify points with *concentrated* tracing effects, we use a heuristic for filtering points. Given the output of Causal Tracing analysis for a point, i.e. one effect per layer (the max across tokens), we define the point to have concentrated tracing effects when there are no more than three layers that have at least 50% of the maximum effect across layers (besides the layer with the max effect itself). Under this criterion, about 10% of the data (74 of 652 cases) have concentrated effects. Note we use our default tracing window size of 5 with the 28 layer GPT-J model for this experiment.

We show the results from our analysis on this data subset in Table 2, and **we observe no changes** in our main conclusions. For ROME with Error Injection, the added effect is 0.2%. Across editing problems and edit methods, the maximum added effect of including tracing effects on R^2 values for predicting rewrite score remains at 3.2% (for Fact Forcing with constrained finetuning). Thus, we conclude that even when facts appear to be stored in a small number of layers, localization results from Causal Tracing are still not informative about editing success, while the choice of edit layer is a far more important factor in whether a fact is successfully edited.

Measuring essence drift. Meng et al. [21] describe one possible consequence of model editing as *essence drift*, which occurs when core properties of an entity change after attempting to edit only one property of that entity. For example, changing where an island is located might also cause the model to nonsensically treat the island as a university campus (see example in Meng et al. [21]).

We aim to obtain an automatic metric to serve as a rough proxy for essence drift. A related metric is calculated with "Local Neutral" data involving the same subject entity but with other properties that are logically neutral with the original property of the subject being edited [15]. However, we do not have "Local Neutral" data for the CounterFact dataset, and essence drift aims to specifically measure changes to *core* properties of a subject.

Therefore, we automatically estimate changes to known properties of the subject s by calculating the change in model perplexity over samples of text that were drawn from the pre-edit model given the prompt "s is a " (which tend to describe a number of key properties of the subject s). We term these samples *essence texts*, and we obtain five samples per subject prompt by sampling with multinomial top-k sampling using $k = 5$. Given our essence texts, we measure the perplexity over the samples before and after editing a fact in the model, for every edited fact in our dataset. Note this is quite similar to the essence drift regularization objective used in the ROME optimization objective [21],

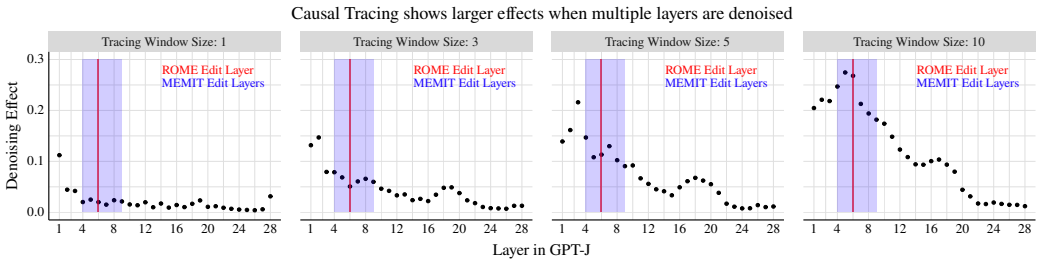


Figure 10: Tracing effects grow larger as the number of adjacent restored layer representations increases (tracing window size).

Rewrite Score Table		R^2 Values				
Editing Problem	Method	Layer	Trace	Both	Diff	p-value
Error Injection	FT (1 layer)	0.756	0.062	0.758	0.002	<1e-4
	FT (5 layers)	0.775	0.055	0.777	0.002	<1e-4
	ROME (1 layer)	0.947	0.016	0.948	0.001	<1e-4
	MEMIT (5 layers)	0.677	0.024	0.678	0.001	0.199
Tracing Reversal	FT (1 layer)	0.067	0	0.067	0	0.997
	FT (5 layers)	0.751	0.045	0.752	0.001	0.032
	ROME (1 layer)	0.294	0.017	0.31	0.015	<1e-4
	MEMIT (5 layers)	0.212	0.036	0.218	0.006	<1e-4
Fact Erasure	FT (1 layer)	0.643	0.028	0.646	0.003	<1e-4
	FT (5 layers)	0.698	0.025	0.70	0.002	<1e-4
	ROME (1 layer)	0.857	0.019	0.858	0	0.555
	MEMIT (5 layers)	0.925	0.019	0.925	0	0.669
Fact Amplification	FT (1 layer)	0.383	0.014	0.393	0.01	<1e-4
	FT (5 layers)	0.424	0.01	0.436	0.011	<1e-4
	ROME (1 layer)	0.88	0.02	0.88	0	0.654
	MEMIT (5 layers)	0.905	0.018	0.906	0.001	<1e-4
Fact Forcing	FT (1 layer)	0.697	0.104	0.724	**0.027**	<1e-4
	FT (5 layers)	0.634	0.10	0.666	**0.032**	<1e-4
	ROME (1 layer)	0.422	0.004	0.425	0.003	<1e-4
	MEMIT (5 layers)	0.345	0.041	0.354	0.009	<1e-4

Table 3: R^2 values for predicting **rewrite** score from choice of edit layer and tracing effect, across editing problem variants (corresponds to data in Fig. 6). Diff shows the added effect of including tracing in the regression (Both vs. Layer Only), in terms of R^2, and p-value shows the results from an F-test comparing the Both and Layer Only models. Tracing has some predictive value for Fact Forcing, but the R^2 value remains small compared to the choice of edit layer.

but we consider it as a metric here. We scale the change in perplexity to a fraction of 5, with the cut-off of 5 chosen to represent a maximally bad change to the model perplexity. Similar to our other metrics, our essence score is 1 if model perplexity on the essence texts does not change after editing the model (capping to 1 in cases of slight decreases in perplexity), and it is 0 if the perplexity increases by 5 or more.

We show essence scores for editing methods across layers in 8. Interestingly, the trend across layers for this metric is mostly counter to the trends for other metrics (Fig. 7), with editing later layers being generally preferable to editing earlier layers. As a result, when combined with the other metrics in Fig. 9, we see that the overall score trend flattens and shifts slightly toward mid-range layers in the model.

C Robustness Experiments

In addition to our main results with ROME for GPT-J and our Rewrite Score metric, we include robustness experiments to confirm that results are similar for (1) other measures of edit success including Paraphrase Score, Neighborhood Score, and Overall Score (Tables 4, 5, and 6), (2) different

Table 4: R^2 values for predicting **paraphrase** score from choice of edit layer and tracing effect, across editing problem variants. Diff shows the added effect of including tracing in the regression (Both vs. Layer Only), in terms of R^2, and p-value shows the results from an F-test comparing the Both and Layer Only models. The added effect of including tracing effects is very small across conditions (less than 3%).

Paraphrase Score Table		R^2 Values				
Editing Problem	Method	Layer	Trace	Both	Diff	p-value
Error Injection	FT (1 layer)	0.061	0.005	0.063	0.002	0.258
	FT (5 layers)	0.036	0.003	0.038	0.001	0.582
	ROME (1 layer)	0.279	0.001	0.303	0.024	<1e-4
	MEMIT (5 layers)	0.246	0	0.269	0.023	<1e-4
Tracing Reversal	FT (1 layer)	0.004	0.001	0.004	0	0.989
	FT (5 layers)	0.001	0	0.002	0.001	0.841
	ROME (1 layer)	0.01	0	0.012	0.002	0.121
	MEMIT (5 layers)	0.001	0	0.001	0	0.997
Fact Erasure	FT (1 layer)	0.046	0.001	0.048	0.002	0.303
	FT (5 layers)	0.079	0.007	0.084	0.005	0.004
	ROME (1 layer)	0.537	0.012	0.539	0.001	0.218
	MEMIT (5 layers)	0.586	0.015	0.587	0.001	0.184
Fact Amplification	FT (1 layer)	0.005	0.012	0.022	0.017	<1e-4
	FT (5 layers)	0.017	0.013	0.035	0.018	<1e-4
	ROME (1 layer)	0.24	0.002	0.267	0.027	<1e-4
	MEMIT (5 layers)	0.236	0.001	0.263	0.026	<1e-4
Fact Forcing	FT (1 layer)	0.044	0.004	0.046	0.002	0.367
	FT (5 layers)	0.023	0.002	0.025	0.002	0.387
	ROME (1 layer)	0.357	0.01	0.36	0.003	0.003
	MEMIT (5 layers)	0.095	0.001	0.105	0.01	<1e-4

Table 5: R^2 values for predicting **neighborhood** score from choice of edit layer and tracing effect, across editing problem variants. Diff shows the added effect of including tracing in the regression (Both vs. Layer Only), in terms of R^2, and p-value shows the results from an F-test comparing the Both and Layer Only models. The added effect of including tracing effects is very small across conditions (2% or less).

Neighborhood Score Table		R^2 Values				
Editing Problem	Method	Layer	Trace	Both	Diff	p-value
Error Injection	FT (1 layer)	0.005	0	0.008	0.002	0.197
	FT (5 layers)	0.014	0.001	0.015	0.001	0.55
	ROME (1 layer)	0.011	0.003	0.015	0.005	0.001
	MEMIT (5 layers)	0.004	0.001	0.006	0.002	0.154
Tracing Reversal	FT (1 layer)	0.001	0	0.001	0	1
	FT (5 layers)	0.001	0	0.002	0.001	0.946
	ROME (1 layer)	0.001	0	0.002	0.001	0.946
	MEMIT (5 layers)	0.001	0	0.002	0	0.981
Fact Erasure	FT (1 layer)	0.01	0	0.014	0.004	0.037
	FT (5 layers)	0.01	0	0.013	0.004	0.06
	ROME (1 layer)	0.04	0.005	0.046	0.006	0.001
	MEMIT (5 layers)	0.05	0.007	0.059	0.009	<1e-4
Fact Amplification	FT (1 layer)	0.012	0.009	0.02	0.008	<1e-4
	FT (5 layers)	0.016	0.008	0.025	0.009	<1e-4
	ROME (1 layer)	0.04	0.01	0.05	0.01	<1e-4
	MEMIT (5 layers)	0.035	0.008	0.044	0.01	<1e-4
Fact Forcing	FT (1 layer)	0.054	0	0.057	0.003	0.03
	FT (5 layers)	0.019	0.001	0.022	0.004	0.011
	ROME (1 layer)	0.299	0.022	0.311	0.012	<1e-4
	MEMIT (5 layers)	0.046	0.012	0.066	0.02	<1e-4